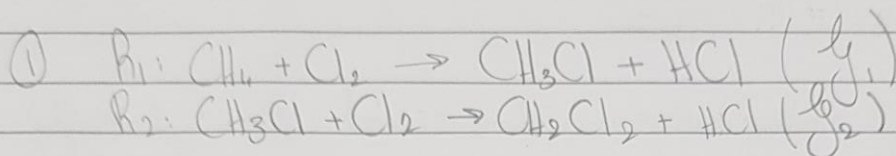


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CHE 210 HW 8

② $DOF = (4a + 4b) - (5a + 5b + 5c + 5d)$
 overall = $(16 + 2) - (1 + 1 + 1 + 15) = 0$, specified
 Reactor = $(6 + 2) - (1 + 1 + 1 + 5) = 0$, specified
 Condenser = $(8 + 0) - (0 + 0 + 0 + 4) = 4$, underspecified
 Absorber = $(4 + 0) - (0 + 0 + 0 + 2) = 2$, underspecified
 Still = $(4 + 0) - (0 + 0 + 0 + 2) = 2$, underspecified
 Mixing Point = $(5 + 0) - (1 + 1 + 0 + 2) = 1$, underspecified

b) Species balance on mixing point

$CH_4: \dot{n}_1 + \dot{n}_3 = 80 \text{ kmol } CH_4/h \Rightarrow \dot{n}_1 = 80 - \dot{n}_3 \Rightarrow \dot{n}_3 = 80 - \dot{n}_1$
 $Cl_2: \dot{n}_2 = 20 \text{ kmol } Cl_2/h$

Atomic balance on reactor

C: $80(1) = (1)\dot{n}_3 + (1)5\dot{n}_4 + (1)\dot{n}_5 \rightarrow \dot{n}_3 + 5\dot{n}_4 = 80 \text{ kmol/h}$
 H: $4(\dot{n}_3) + \dot{n}_4 + 17\dot{n}_5 = 4(80) \text{ kmol/h}$
 Cl: $\dot{n}_2 + (7)\dot{n}_5 = 2(20) \text{ kmol/h}$

$\boxed{\dot{n}_3 = 62.9 \text{ kmol/h}, \dot{n}_4 = 20 \text{ kmol/h}, \dot{n}_5 = 2.86 \text{ kmol/h}}$

① Atomic balance on reactor

$$C: \dot{n}_1 + \dot{n}_3 = \dot{n}_3 + 6\dot{n}_5$$

$$H: 4\dot{n}_1 = 4\dot{n}_3 + \dot{n}_4 + 17\dot{n}_5 \Rightarrow 17\dot{n}_5 = 4\dot{n}_1 - 4\dot{n}_3 - \dot{n}_4$$

$$Cl: 2\dot{n}_2 = \dot{n}_4 + 7\dot{n}_5 \longrightarrow 7\dot{n}_5 = 2\dot{n}_2 - \dot{n}_4$$

$$5\dot{n}_5 = \dot{n}_5 / M_{CH_3Cl} = 1000 \text{ g/h} / (50.46) = 19.8 \text{ kmol/h}$$

$$5\dot{n}_5 = 19.8 \text{ kmol/h} \longrightarrow \boxed{\dot{n}_5 = 3.96 \text{ kmol/h}}$$

$$67.32 \text{ kmol/h} = 4\dot{n}_1 - 4\dot{n}_3 - \dot{n}_4 \Rightarrow \boxed{\dot{n}_1 = 23.76 \text{ kmol/h}}$$

$$27.72 \text{ kmol/h} = 2\dot{n}_2 - \dot{n}_4$$

$$\dot{n}_1 + \dot{n}_3 = 4\dot{n}_2$$



$$\begin{aligned} \dot{n}_1 &= 23.76 \text{ kmol/h} \\ \dot{n}_2 &= 8.36 \text{ kmol/h} \\ \dot{n}_3 &= 9.68 \text{ kmol/h} \end{aligned}$$

$$\dot{n}_{\text{feed}} = \dot{n}_1 + \dot{n}_2 = 32.13 \text{ kmol/hr}$$

$$X_{CH_4, \text{feed}} = 23.76 / 32.13 = 74\%$$

$$X_{Cl_2, \text{feed}} = 100\% - 74\% = 26\%$$



$$\dot{n}_{\text{recycle}} = \dot{n}_3 = 9.68 \text{ kmol/h}, 100\% \text{ } CH_4 \text{ only}$$

② Basis: 100 g/mol/min

Species balance on Mixer

on distiller 1

$$\text{CH}_4: 70 + X_{\text{CH}_4,i}(100) = X_{\text{CH}_4,F}(100)$$

$$\text{C}_2\text{H}_6: 30 + X_{\text{C}_2\text{H}_6,i}(100) = X_{\text{C}_2\text{H}_6,F}(100)$$

$$X_{\text{CH}_4} + X_{\text{C}_2\text{H}_6} = 1$$

$$X_{\text{CH}_4,F}(100) = 0.9\dot{n}_2 + 0.02\dot{n}_3$$

$$X_{\text{C}_2\text{H}_6,F}(100) = 0.1\dot{n}_2 + 0.08\dot{n}_3$$

on distiller 2

$$\text{CH}_4: 0.9\dot{n}_2 = 0.97\dot{n}_4 + X_{\text{CH}_4}(100)$$

$$\text{C}_2\text{H}_6: 0.1\dot{n}_2 = 0.03\dot{n}_4 + X_{\text{C}_2\text{H}_6}(100)$$

$$X_{\text{C}_2\text{H}_6} + X_{\text{CH}_4} = 1$$

Overall balance

$$(70 = 0.97\dot{n}_4 + 0.02\dot{n}_3) \times 0.6 \Rightarrow 3420 = 47.58\dot{n}_4 + 0.02\dot{n}_3$$

$$30 = 0.03\dot{n}_4 + 0.08\dot{n}_3$$

$$\begin{aligned} 3420 &= 47.58\dot{n}_4 + 0.02\dot{n}_3 \\ - 30 &= 0.03\dot{n}_4 + 0.08\dot{n}_3 \\ \hline 3400 &= 47.50\dot{n}_4 \end{aligned}$$

$$70 = 0.97(71.58) + 0.02\dot{n}_3$$

$$\text{final product} \Rightarrow \dot{n}_4 = 71.58 \text{ g/mol/min}$$

$$\dot{n}_3 = 28.87 \text{ g/mol/min}$$

$$\text{Total mass of distiller 1: } \dot{n}_1 = \dot{n}_3 + \dot{n}_2 \Rightarrow \dot{n}_2 = 171.63 \text{ g/mol/min}$$

③

$$\text{CH}_4: 0.9(171.63) = 154.467 \text{ g/mol CH}_4/\text{min}$$

$$\text{C}_2\text{H}_6: 0.1(171.63) = 17.163 \text{ g/mol C}_2\text{H}_6/\text{min}$$

Species balance on distiller 2

$$\text{CH}_4: 154.467 = 0.97(71.58) + X_{\text{CH}_4}(100)$$

$X_{\text{CH}_4} = 85\%$ recycled stream, $X_{\text{C}_2\text{H}_6} = 15\%$ composition
if species balance fraction recoveries

$$\text{CH}_4: 0.95(100) + 70 = X_{\text{CH}_4}(200)$$

$$\text{CH}_4: \frac{X_{\text{CH}_4, 4\dot{n}_4}}{X_{\text{CH}_4, \dot{n}_1}} = \frac{0.97(71.58)}{0.7(100)} = 99.2\%$$

$$\text{feed stream } X_{\text{CH}_4} = 0.775$$

$$\text{composition } X_{\text{C}_2\text{H}_6} = 0.225$$

$$\text{C}_2\text{H}_6: \frac{X_{\text{C}_2\text{H}_6, 3\dot{n}_3}}{X_{\text{C}_2\text{H}_6, \dot{n}_1}} = 92.67\%$$

③ Basis: 100 g/mol/min

$$0.3(100) = 30 = Y_{C_2H_6}(50) + X_{C_2H_6}(50)$$

$$Y_{C_2H_6} + Y_{H_2O} = 1$$

$$0.7(100) = 70 = Y_{H_2O}(50) + X_{H_2O}(50)$$

$$X_{C_2H_6} + X_{H_2O} = 1$$

When $T = 85.3^\circ\text{C}$

$$30 = 0.4704(50) + 0.1288(50)$$
$$30 \approx 29.071$$

$$f_{A, C_2H_6}(V) = \frac{Y_{C_2H_6} \text{ vapor}}{Z_{C_2H_6} \text{ in f}} = \frac{0.47(50)}{0.3(100)} = 0.783$$

$$f_{B, H_2O}(L) = \frac{X_{H_2O} \text{ liquid}}{Z_{H_2O} \text{ in f}} = \frac{0.876(50)}{0.7(100)} = 0.626$$

$$\Delta_{C_2H_6-H_2O} = \frac{Y_{C_2H_6}}{X_{C_2H_6}} \cdot \frac{Y_{H_2O}}{X_{H_2O}} = \left(\frac{0.47}{0.124} \right) \left(\frac{0.876}{0.53} \right)$$
$$= 6.26$$